

Motor Selection Assignment



ME30295: Electronics, Signals and Drives

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Summary

This report details the process for selecting a suitable electric motor to drive a belt conveyor used in automated packaging of laser printers. Using data provided about the physical characteristics of the belt conveyor, it was possible to calculate the relevant torque values needed to select a motor capable of handling the required load.

The Maxon-Parvalux EPM63-100 motor was selected as the most suitable choice for the application. The EPM63-75 was also checked as a step down in size range, but this was found to not fulfil the needs of the belt conveyor. The Maxon-ESCON 70/10 was selected as the most appropriate converter for the task. A two-stage belt pulley with overall ratio 21.77 was used to reduce the angular velocity between the motor and the belt conveyor.

In order to meet increasing demand for laser printers, possible optimisations in the process were then analysed. Using the drive system selected earlier, the velocity profile could be recalculated to show that the fastest operation of the machine is 3.8 seconds per cycle while staying within motor peak and r.m.s torque limits. Implementation of a 'filling on the move' velocity profile where packing can be done at both 10 % and 20 % of the maximum velocity was further found to reduce the cycle time to 3.54 and 3.29 seconds, respectively.

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1. Introduction

This report details the process for selecting a suitable electric motor to drive a belt conveyor used in automated packaging of laser printers. The purpose of this system is to move empty boxes from an Infeed Area to a Packing Station where the product is placed in the box, before finally moving the filled box to the Despatch Area where the boxed product can be shipped.

Section 2 of this document details the process of selecting a motor to drive the belt conveyor while Section 3 details possible optimisations of the system to allow a faster cycle time and therefore increase how fast products can be processed.

2. Motor Selection

Figure 1 shows the mechanical schematic of the belt conveyor system. A single electric motor drives a 2-stage belt pulley transmission which is connected to a driving roller of the belt conveyor. The belt conveyor runs over 2 larger pulleys (a driving and a driven pulley) and there is a third tensioning pulley shown at the bottom of Figure 1. The motor is connected to a converter which provides a trapezoidal velocity profile which first moves the empty boxes from the Infeed Area to the Packing Area where the boxes stop and are filled, and then finally to the Despatch Area where they are removed.

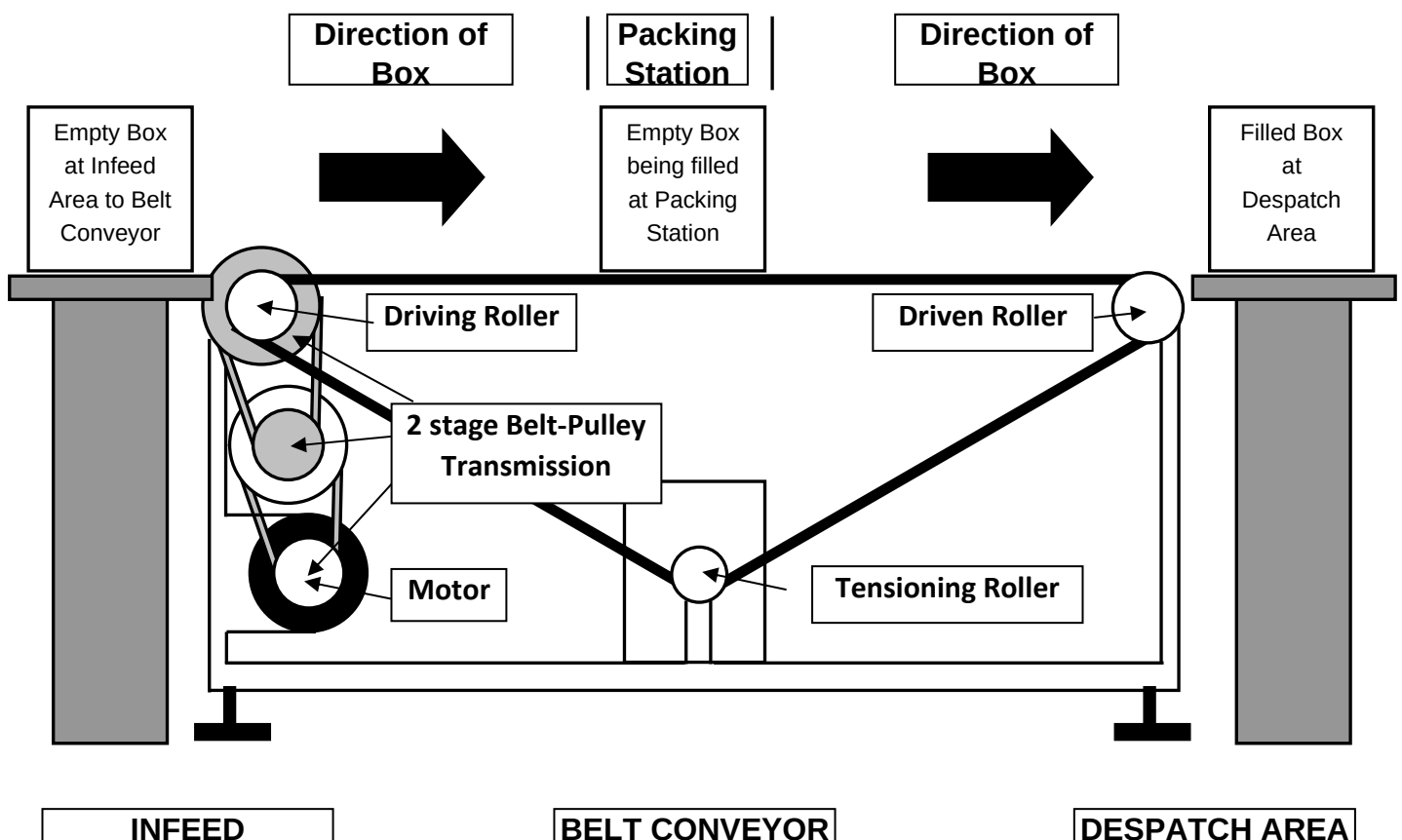


Figure 1 Mechanical system diagram for the belt conveyor with motor and 2-stage belt transmission shown

Length of Belt Conveyor:	3 m	
Position of Packing Station:	1.5 m	(centrally placed along belt conveyor's length)
Weight of Belt Conveyor:	250 N	
Belt Conveyor Friction Force:	250 N	(measured at the conveyor belt)
Number of Rollers:	3	(driving, driven and tensioning roller)
Diameter / Weight of a Main Roller:	0.15 m / 35 N	(rollers of a hollow construction)
Diameter / Weight of the Tension Roller:	0.1 m / 20 N	(rollers of a hollow construction)
Weight of Empty Box:	4 N	
Weight of Product:	90 N	
Box Filling Time:	2 s	(during which time belt conveyor is stationary)
Maximum Belt Velocity	1.0 m s ⁻¹	

Figure 2 Physical characteristics of the belt conveyor system and required velocity profile

Figure 2 shows the mechanical characteristics of the belt conveyor system as well as details required to define the velocity profile driving the motor.

A brushed PM DC servo motor can be used as it can be controlled to have accurate accelerations and decelerations according to the specified velocity profile, in order to get the box to accurately move the right amount of distance between the stations on the belt conveyor.

2.1. Assumptions

- The centres of the pulleys of the 2-stage belt transmission are separated by 300mm.
- The 2 large pulleys and 2 small pulleys used to make the 2-stage transmission are the same – giving both stages the same transmission ratio.
- Running mechanical losses throughout the system can be estimated at 50 W total.
- The inertia of the belt conveyor and any boxes on it can be considered to act on the circumference of the Driving Roller.
- The rollers of the belt conveyor are hollow and can be approximated as thin walled tubes with inertia: $I = mr^2$ kgm² (m = mass, r = radius).
- The inertia of each belt in the transmission acts on their respective small pulleys within the transmission.
- Inertias of minor components such as bearings and connecting shafts can be neglected as these are of the order of 10⁻⁵ kgm².

2.2. Calculations

2.2.1. Angular roller velocity

For a linear conveyor belt velocity of 1 m/s and a driving roller radius of 0.075 m the angular velocity of the driving and driven rollers is:

$$\omega_{\text{driving}} = \frac{1}{0.075} = \mathbf{13.33 \text{ rad/s} = 127.3 \text{ rpm}}$$

2.2.2. Power supplied by the belt

Continuous power supplied by the conveyor belt can be found using the frictional forces (250 N) and the constant running velocity (1 m/s):

$$P_{\text{cont}} = Fv = 250 \cdot 1 = 250 \text{ W}$$

Adding 50 W for estimated constant mechanical losses in the system:

$$P_{\text{cont}} = \mathbf{300 \text{ W}}$$

2.2.3. Torque at roller

Using the previous calculated power and angular speed values, the torque at the driving roller can be calculated:

$$T_{\text{driving}} = \frac{P_{\text{cont}}}{\omega_{\text{driving}}} = \frac{300}{13.33} = \mathbf{22.5 \text{ Nm}}$$

2.2.4. Roller inertias

Modelling the rollers as thin walled tubes the inertias for the main rollers and tension roller can be approximated as:

$$J_{\text{main}} = \left(\frac{35}{9.81} \right) (0.075)^2 = \mathbf{2.007 \times 10^{-2} \text{ kgm}^2}$$

$$J_{\text{tensioner}} = mr^2 = \left(\frac{20}{9.81} \right) (0.05)^2 = \mathbf{5.097 \times 10^{-3} \text{ kgm}^2}$$

2.2.5. Belt and boxes inertia

As assumed previously, the weight of the conveyor belt (250 N) and 2 boxes (one filled – 94 N, one empty – 4 N) can be considered to act at the circumference of the driving roller. Hence the inertia for these components is:

$$J_{\text{belt+boxes}} = mr^2 = \left(\frac{348}{9.81} \right) (0.075)^2 = \mathbf{0.1997 \text{ kgm}^2}$$

2.2.6. Gear ratios

Assuming a motor speed of 3000 rpm (the maximum for motors of type and approximate size as supplied by Maxon), an initial motor speed estimation can be made using the driving roller angular speed calculated in Section 2.2.1:

$$G. R = \frac{3000}{127.3} = \mathbf{23.57}$$

Since a two-stage transmission is going to be used the square root of this gives **4.85** as the ratio for each stage of the transmission (assuming both parts are identical). This is the ideal gear ratio that would allow the motor to run at 3000 rpm when the conveyor is moving at 1 m/s. However, it may not be possible to match this due to limitations on belt and pulley parts. So long as the overall transmission ratio is at or below 23.57 then the motor will be able to run at or below 3000 rpm.

2.2.7. Selecting belts and pulleys

Using the SDI/SP online centre distance designer it was possible to generate a belt and pulley combination to a centre distance of approximately 300 mm and a gear ratio of just under 4.85 (**4.667**) – see corresponding screen capture Figure 3.

Figure 3 Screen capture of the SDI/SP website centre distance belt drive generator tool with important values highlighted

The components selected are:

- **A 6R25M190150**: neoprene single-sided belt (190 Teeth, 15mm wide, 5 mm pitch)
- **A 6C24M024TM20**: Steel pulley (**small** – 24 Teeth, 5mm Pitch)
- **A 6C24M112TM15**: Steel pulley (**large** – 112 Teeth, 5mm Pitch)

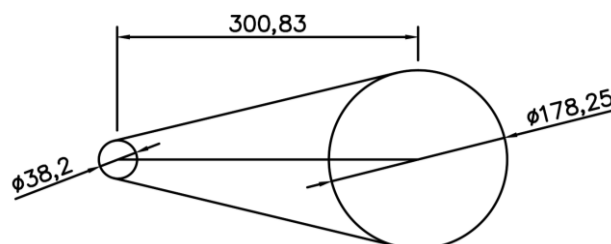


Figure 4 Diagram of pulley size and centre distance

Through a 2 stage transmission the overall gear ratio is **21.77**. This is slightly lower than the ideal 23.57 required to run at 3000 rpm but this is preferable to going over it as the motor can simply be supplied less voltage to achieve the same conveyor belt speed for a lower gear ratio.

The maximum angular velocity of the motor using this transmission ratio and the driving roller angular velocity is:

$$\omega_{\text{motor}} = 127.3 \times 21.77 = \mathbf{2771 \text{ rpm} = 290.18 \text{ rad/s}}$$

The inertia of the pulleys in the transmission has been calculated using CAD models provided on the SDI/SP website (Figure 5). By setting these 3D models to have a density of 7.85 g/cm³ (steel) an accurate calculation of their rotational inertia is given. These models include a 25 mm diameter steel shaft to attach the pulleys to bearings/motors.

The belt drive attaching the two pulleys can be approximated as a thin ring, so the inertia is **mr²**. Approximating the mass as 0.01 kg and average radius as 0.1 m the inertia is 2.25 x 10⁻⁴ kgm².

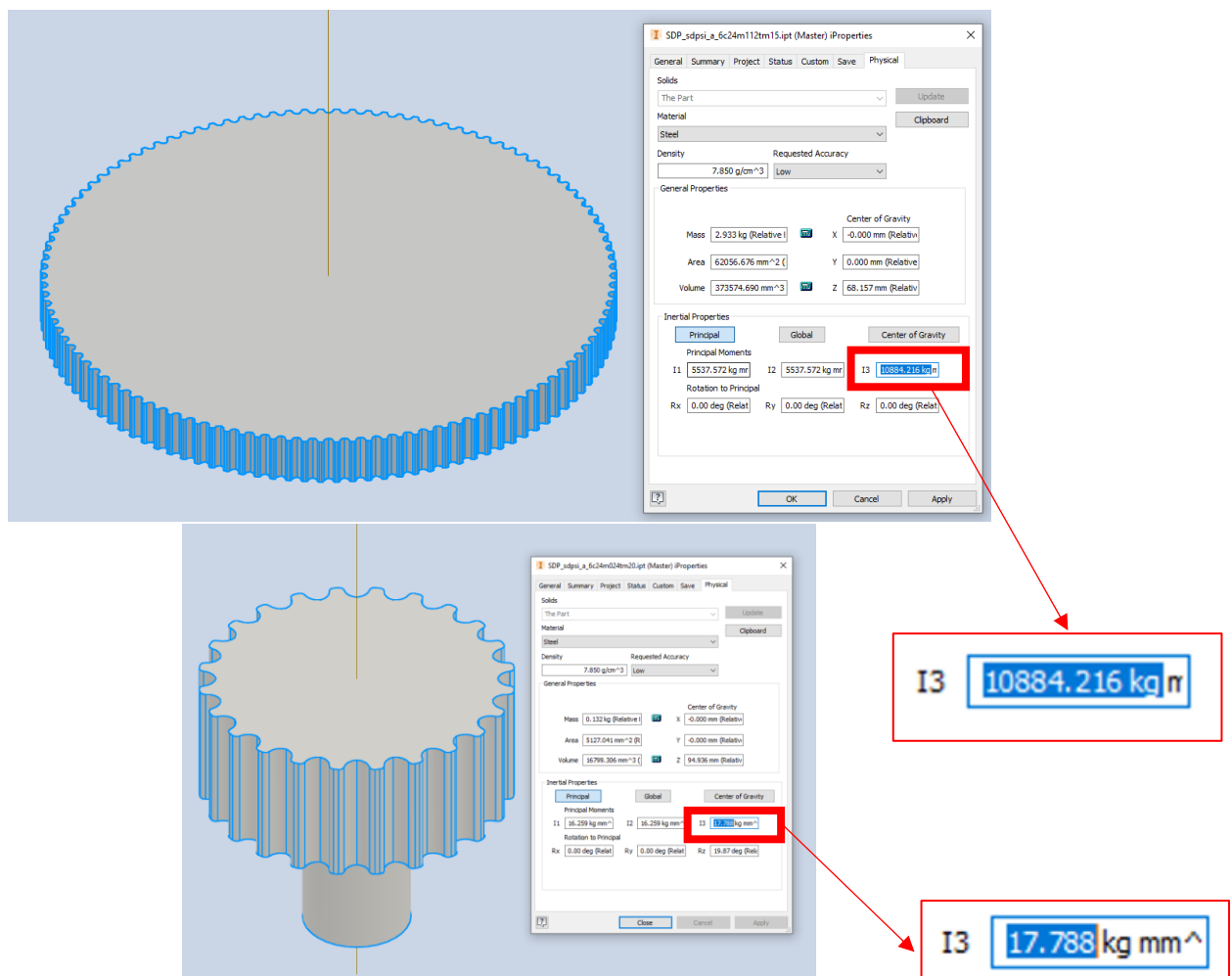


Figure 5 Inertia calculations for the pulleys in the belt drive transmission. Converting the inertia values from kgmm² to kgm² gives: Large: **1.088 x 10⁻² kgm²**. Small: **1.779 x 10⁻⁵ kgm²**

2.2.8. Motor torque

The constant motor torque can be found using the values calculated for motor speed (2771 rpm) and maximum power (300 W):

$$T_{\text{const}} = \frac{P_{\text{cont}}}{\omega_{\text{motor}}} = \frac{300}{2771 \cdot \frac{2\pi}{60}} = \mathbf{1.034 \text{ Nm}}$$

2.3. Initial Motor Selection

Maxon-parvalux EPM63-100 (www.parvalux.com/product/epm63-100-vented-48v-3000-rpm) is the initial selection to meet the parameters calculated in the previous section. It has a rated speed of 3000 rpm, 471 W max power, a constant torque of 1.5 Nm and inertia of **9.64 x 10⁻⁴ kgm²**.

2.4. Grouping Inertias

From the perspective of the motor different groups of components have different inertias based on how many gear ratios the motor experiences the inertia through. The 2 stage belt transmission means that there are 3 groups which must be accounted for:

$$\text{Motor Inertia (1)} = J_{\text{motor}} + J_{\text{belt 1}} + J_{\text{pulley 1}} = 0.0012 \text{ kgm}^2$$

$$\text{Transmission Inertia (2)} = J_{\text{belt 2}} + J_{\text{pulley 2}} + J_{\text{pulley 1}} = 0.01112 \text{ kgm}^2$$

$$\text{Belt Inertia (3)} = J_{\text{belt+boxes}} + 2J_{\text{main roller}} + J_{\text{tensioner}} + J_{\text{pulley 2}} = 0.2558 \text{ kgm}^2$$

These have been calculated using the values calculated previously. Pulleys 1 and 2 are the small and large pulleys respectively.

To account for the motor experiencing the inertias through the transmission the transmission group must be divided by the gear ratio of the first stage squared, while the belt stage must be divided by the gear ratio (of one stage) to the fourth power:

$$J_{\text{Total}} = J_1 + \frac{J_2}{4.85^2} + \frac{J_3}{4.85^4} = \mathbf{2.142 \times 10^{-3} \text{ kgm}^2}$$

2.5. Initial Velocity Profile

The Packing Station is located centrally on the conveyor; 1.5m from either end. Each box must be accelerated from the Infeed area to the maximum 1 m/s belt speed, before being decelerated to stop at the Packing Station. This cycle is then repeated to move the filled box to the Despatch Area. One cycle is shown in Figure 6. In order to move the box between the stations the area under the velocity profile must be 1.5 m. As specified, the box must wait at the Packing Station for 2 seconds in order to load the product into the box.

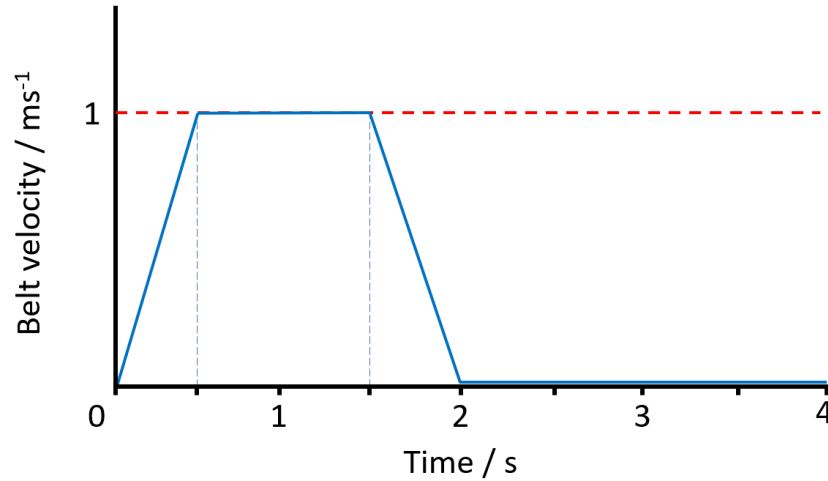


Figure 6 Initial velocity profile for moving a box from one station to another

$$\text{Area} = 2 \left(\frac{0.5(1)}{2} \right) + 1(1) = \mathbf{1.5 \text{ m}}$$

The box accelerates from $t = 0 - 0.5 \text{ s}$ up to a velocity of 1 m/s . After 1 second at this speed it decelerates from $t = 1.5 - 2 \text{ s}$ to a stop. It then waits at the Packing Station for 2 seconds while the product is loaded. The cycle is then repeated to move the box to the Despatch Area.

The initial cycle length is therefore **4 seconds**.

2.5.1. Torque in each phase

Torque for the acceleration, running, deceleration and stopped phase can be found with the following equation:

$$T = \frac{T_F}{R} + J_{\text{Total}} \left(\frac{\Delta\omega}{\Delta t} \right)$$

Where T_F is the frictional torque at the driving roller:

$$T_F = 250(0.075) = 18.75$$

$\Delta\omega$ is the change in angular speed (rad/s) across the time interval Δt and R is the overall gear ratio (21.77). Using these values the torque (Nm) in each phase is given in Table 1:

Table 1 Summary of demanded motor torque during each phase of the duty cycle when using motor EPM63-100

T_1 (acceleration)	2.10
T_2 (running)	0.86
T_3 (deceleration)	-0.38
T_4 (stopped)	0.0
T (r.m.s)	0.87

2.6. Motor Suitability Check

Nominal continuous torque	1.5 Nm
Max intermittent torque	2.4 Nm

Figure 7 EPM63-100 continuous and intermittent maximum torque ratings

2.6.1. Specification comparison

The motor rated torques (Figure 7) can then be checked against the predicted demanded torques of the application.

- “Is peak torque $\leq 4 \times \text{r.m.s torque}$?”

$$2.10 \leq 4(0.87) \rightarrow 2.10 \leq 3.48$$

PASSES

- “Is required peak torque < motor peak torque?”

$$2.10 < 2.4$$

PASSES

- “Is required r.m.s torque < motor constant torque?”

$$0.87 < 1.5$$

PASSES

2.6.2. Checking next smallest size

The motor EPM 63-100 is big enough to complete the task. However, it would be more cost effective to use a smaller motor. The next size down in the Parvalux (Maxon) range is the EPM 63-75. This runs at 3000 rpm, but it is rated to 377 W and has $J_{\text{motor}} = 8.53 \times 10^{-4} \text{ kgm}^2$.

Nominal continuous torque	1.2 Nm
Max intermittent torque	1.8 Nm

Figure 8 EPM63-75 continuous and intermittent maximum torque ratings

By replacing J_{motor} with the new value, the torque in each phase could be calculated again as done in Section 2.5.1.

Table 2 Summary of demanded motor torque during each phase of the duty cycle when using motor EPM63-75

T_1 (acceleration)	2.04
T_2 (running)	0.86
T_3 (deceleration)	-0.32
T_4 (stopped)	0.0
T (r.m.s)	0.8475

Using the torque values in Table 2 and the specified motor maximum torques in Figure 8 the motor can be checked as before in Section 2.6.1.

- “Is peak torque $\leq 4 \times \text{r.m.s torque?}$ ”

$$2.04 \leq 4(0.84) \rightarrow 2.04 \leq 3.39$$

PASSES

- “Is required peak torque $<$ motor peak torque?”

$$2.04 < 1.8$$

FAILS

- “Is required r.m.s torque $<$ motor constant torque?”

$$0.8475 < 1.2$$

PASSES

This motor is not suitable for the application since the required peak torque is higher than the motors rated peak intermittent torque.

2.7. Final Motor Selection

The **Maxon-Parvalux EPM63-100 motor** has been selected for this task. The converter to be used is the **Maxon-ESCON 70/10**.

These are available from the following links:

www.parvalux.com/product/epm63-100-vented-48v-3000-rpm - Motor

<https://www.maxongroup.com/maxon/view/product/422969> - Converter



3. Process Optimisations

3.1. Optimising the velocity profile

Using the Motor selected in Section 2 the velocity profile can be adjusted so that the motor is running closer to its specified limits in order to decrease the cycle time and therefore make the belt conveyor more efficient.

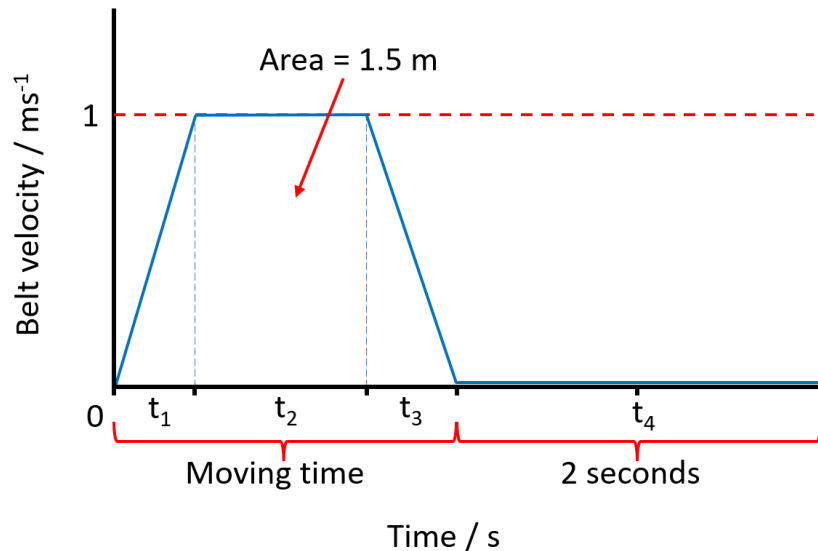


Figure 9 Velocity profile showing 'Moving time' as the dimension to be minimised in order to decrease cycle time

In order to do this the 'Moving time' shown in Figure 9 must be minimised (the 2 second stationary loading period is fixed) while keeping the area under the velocity profile at 1.5 m. This can be done by accelerating and decelerating the conveyor as fast as possible within the limitations of the motor. A spreadsheet (Table 3) was used to iterate to find the shortest possible acceleration and deceleration times (t₁ and t₃). These have been highlighted in bold on the following table (t₁ = 0.405s, t₃ = 0.195s).

Table 3 Iterative table used to find the minimum acceleration and deceleration times within the limits of the selected motor. Conditional formatting used to show where torque during the phases exceeds the rated peak torque of the motor (2.4 Nm)

t ₁ & t ₃ (s)	T ₁ (Nm)	T ₃ (Nm)
0.1	7.08	5.35
0.15	5.01	3.28
0.19	4.13	2.41
0.195	4.05	2.33
0.20	3.97	2.25
0.25	3.35	1.62
0.30	2.93	1.21
0.35	2.64	0.91
0.40	2.42	0.69
0.405	2.396	0.67
0.41	2.38	0.65
0.45	2.24	0.52

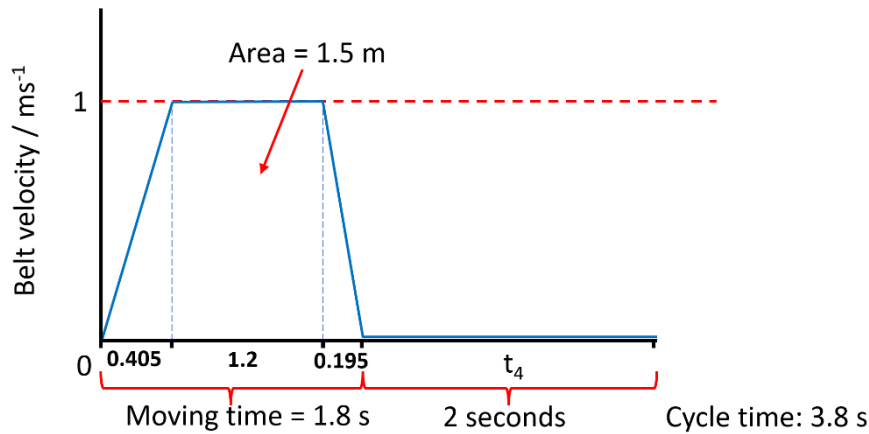


Figure 10 Asymmetric velocity profile using optimised acceleration and decelerations according to motor specifications

$$\text{Area} = \left(\frac{0.405}{2}\right) + \left(\frac{0.195}{2}\right) + 1.2(1) = \mathbf{1.5 \text{ m}}$$

The new cycle time is **3.8s** – a 5% improvement on the original cycle. Using these newly calculated times and torque values the r.m.s torque is **1.06 Nm**.

3.2. 'Filling on the move'

In this case the boxes can move at either 10 % or 20% of the maximum speed through the packing area. This will alter the velocity profile and reduce the cycle time.

3.2.1. Assumptions

- The first cycle can be ignored (where the conveyor starts from stationary) and the conveyor can be considered constantly running.
- Packing the boxes still takes 2 seconds.
- The packing area remains symmetric about the centre (1.5m from either end) of the conveyor.

3.2.2. Optimising velocity profiles

During the 2 second packing process the box will move 0.2 m and 0.3 m respectively for 10 % and 20 % speeds. If the Packing Station is centred at the halfway on the conveyor the velocity profiles will look like Figure 11 (showing 1 cycle which would move a box halfway along).

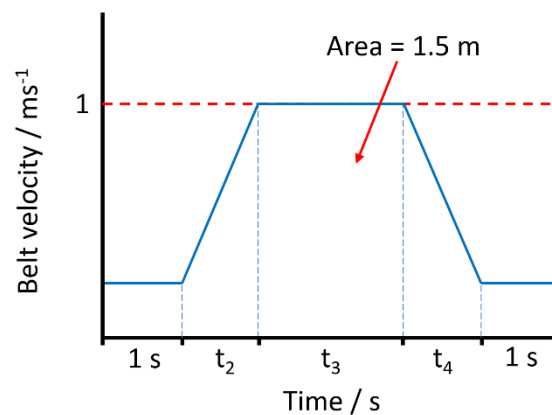


Figure 11 'Filling on the move' velocity profile

The distance moved during filling will be 0.2 m and 0.4 m for 10% and 20% speed respectively. This would leave 1.3m and 1.1m for the remainder of the velocity profile for 10% and 20% respectively.

As before, in Section 3.1, the acceleration and deceleration phases (t_2 and t_4) should be shortened until they are at the limit of the motor specifications (Figure 7), while the running speed section (t_3) should be as long as possible. A similar spreadsheet to Table 3 has been used to find the optimal time periods for both 10% and 20% velocity profiles with the $\Delta\omega$ value modified for each to reflect the increase in baseline speed ($\Delta\omega_{10} = 261.15 \text{ rad/s}$, $\Delta\omega_{20} = 232.13 \text{ rad/s}$).

3.2.2.1. 10 % speed calculations

Table 4 Iterative table used to find the minimum acceleration and deceleration times within the limits of the selected motor. Conditional formatting used to show where torque during the phases exceeds the rated peak torque of the motor (2.4 Nm)

t_2 & t_4 (s)	T_2 (Nm)	T_4 (Nm)
0.1	6.46	4.73
0.15	4.59	2.87
0.16	4.36	2.63
0.17	4.15	2.43
0.175	4.06	2.34
0.18	3.97	2.25
0.2	3.66	1.94
0.25	3.10	1.38
0.3	2.73	1.00
0.35	2.46	0.74
0.36	2.415	0.69
0.365	2.394	0.67

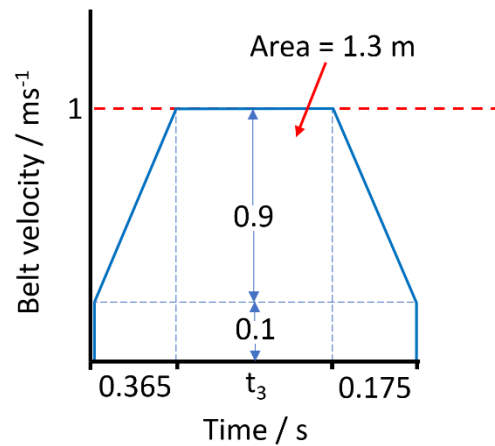


Figure 12 Area diagram for the remaining part of the velocity profile

$$\text{Area}_{t_2+t_4} = \left(\frac{0.365(0.9)}{2} \right) + (0.1)(0.365) + \left(\frac{0.175(0.9)}{2} \right) + (0.1)(0.175) = \mathbf{0.297 \text{ m}}$$

$$\therefore \text{Area}_{t_3} = 1.3 - 0.297 = 1.003 \text{ m} \rightarrow \mathbf{t_3 = 1.003 \text{ s}}$$

$$\mathbf{t_{10\%} = 0.365 + 0.175 + 1.003 + 2 = 3.54 \text{ s}}$$

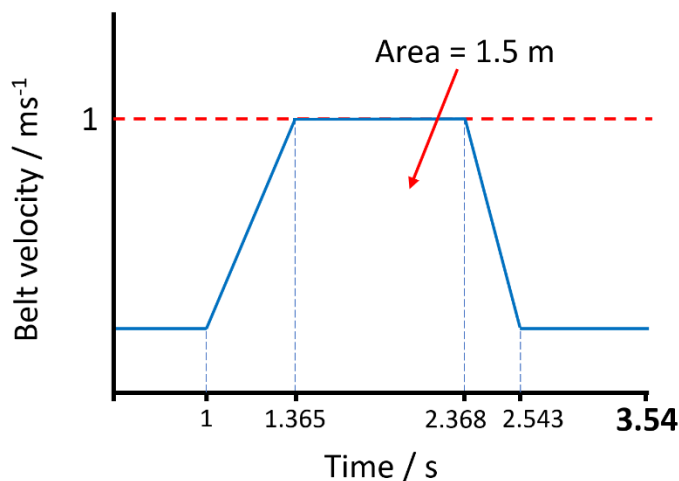


Figure 13 Optimised 10% 'filling on the move' velocity profile

Table 5 Torques (Nm) in each phase of the velocity profile

T_1 (acceleration)	2.39
T_2 (running)	0.861
T_3 (deceleration)	-2.34
T_4 (packing)	0.861
T (r.m.s)	1.22

3.2.2.2. 20 % speed calculations

Table 6 Iterative table used to find the minimum acceleration and deceleration times within the limits of the selected motor. Conditional formatting used to show where torque during the phases exceeds the rated peak torque of the motor (2.4 Nm)

t_2 & t_4 (s)	T_2 (Nm)	T_4 (Nm)
0.1	5.83	4.11
0.15	4.18	2.45
0.155	4.07	2.35
0.16	3.97	2.25
0.2	3.35	1.62
0.25	2.85	1.13
0.3	2.52	0.80
0.31	2.47	0.74
0.32	2.42	0.69
0.325	2.39	0.67

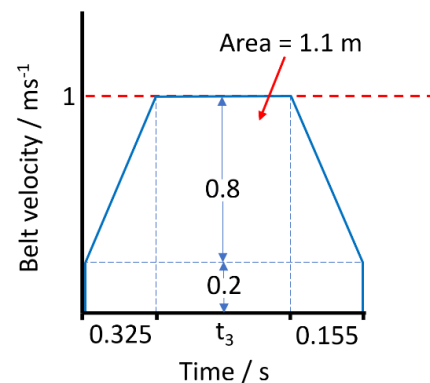


Figure 14 Area diagram for the remaining part of the velocity profile

$$\text{Area}_{t_2+t_4} = \left(\frac{0.325(0.8)}{2} \right) + (0.2)(0.325) + \left(\frac{0.155(0.8)}{2} \right) + (0.2)(0.155) = \mathbf{0.288 \text{ m}}$$

$$\therefore \text{Area}_{t_3} = 1.1 - 0.288 = 0.812 \text{ m} \rightarrow t_3 = \mathbf{0.812 \text{ s}}$$

$$t_{20\%} = 0.325 + 0.155 + 0.812 + 2 = \mathbf{3.29 \text{ s}}$$

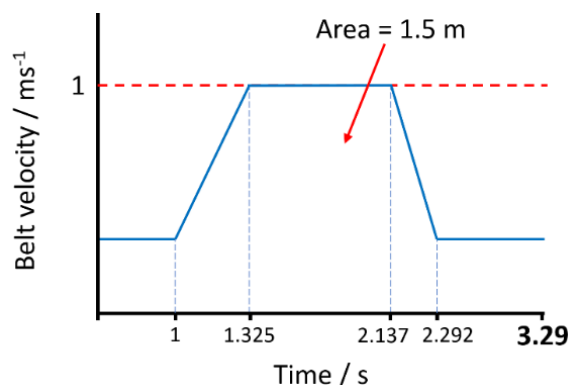


Figure 15 Optimised 20% 'filling on the move' velocity profile

T_1 (acceleration)	2.39
T_2 (running)	0.861
T_3 (deceleration)	-2.35
T_4 (packing)	0.861
T (r.m.s)	1.21

Table 7 Torques (Nm) in each phase of the velocity profile

All torque values calculated for both the 10 % and 20 % 'filling on the move' velocity profiles are within the specified limits given by the manufacturer (Figure 7 – 2.4 Nm peak torque, 1.5 Nm r.m.s torque). These are the most optimised velocity profiles for these parameters as the acceleration and deceleration rates were iterated up to the limit of the motors in order to get the shortest cycle time possible. The final cycle times were **3.54 s** and **3.29 s** for 10 % speed and 20 % speed respectively.

3.3. System Issues Considerations

If the boxes on the conveyor are assumed to be cubes with 30 cm sides with the centre of mass at the geometric centre, then there is a limit to how fast the conveyor can accelerate and decelerate before the boxes tip over due to not being attached to the conveyor. Figure 17 shows a simplified diagram of this.

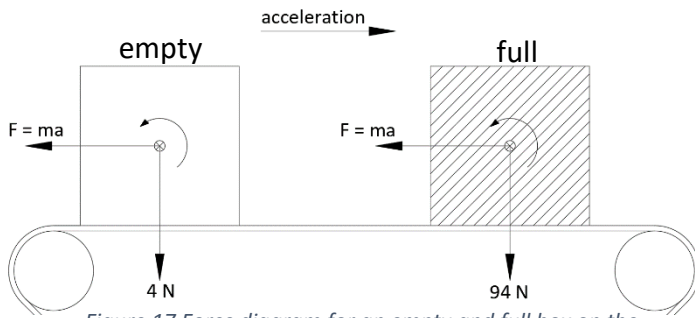


Figure 17 Force diagram for an empty and full box on the conveyor showing forces when accelerating

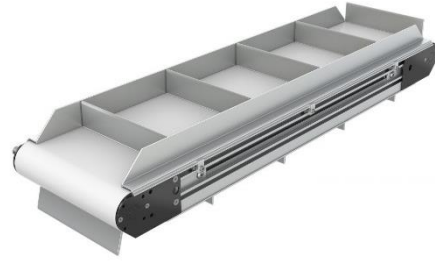


Figure 16 Conveyor featuring 'cleats' to stop boxes from tipping. (Image from: dornerconveyors.com)

The fastest acceleration using the selected motor was found to be -5.13 m/s^2 (decelerating from 1 to 0 m/s in 0.195 s; see Section 3.1). This applies a horizontal force of 2.1 N and 49 N to the empty and full boxes respectively which would create 0.31 Nm and 7.35 Nm of torque acting to tip each box over. This is opposed by the weight of the boxes. In this case the boxes do not tip over. The limit of acceleration due to this effect is $\pm 9.8 \text{ m/s}^2$ at which point the boxes would tip over. This limits how fast the conveyor can be accelerated to 0.102 s. Using the same 21.77 ratio belt transmission exceeding this would require a motor capable of handling intermittent torques of **6.96 Nm**. This motor would be much more expensive, and the incurred cost would likely not be worth it compared to the small time saving. To stop boxes tipping over or moving a conveyor belt with 'cleats' could be introduced (see Figure 16)

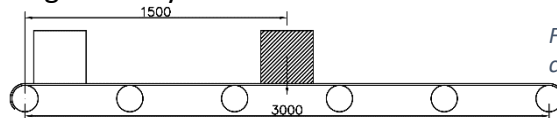


Figure 18 Supporting rollers to stop conveyor sagging at the centre

Another issue shown in Figure 1 is that the conveyor is unsupported across its 3 metre length where the boxes are filled. Figure 18 shows how multiple auxiliary rollers could be used to support the weight of the boxes as they travel along the conveyor to stop it sagging.

4. Conclusion

The **Maxon-Parvalux EPM63-100** has been shown to be a suitable motor for this application. This was done by firstly making a set of initial assumptions and then using these, along with provided data, to perform the necessary calculations and analysis to confirm this motor could meet the requirements of the system. The motor will be coupled with a 2-stage belt and pulley transmission with an overall ratio of **21.77**. Motor control according to the calculated velocity profiles will be provided by the **Maxon-ESCON 70/10** converter.

After initially selecting the motor, it was checked for suitability by comparing the rated peak and r.m.s torques against the predicted demanded torque values from the belt conveyor. The next size down in the Maxon motor range was similarly checked (EPM63-75) and this was confirmed to be too small for the application, meaning EPM63-100 is the motor that should be used.

Finally, optimisations to the system were analysed. Firstly, shortening the acceleration and deceleration phases was considered using an iterative spreadsheet to find the respective shortest times that kept the peak torque within allowable limits. This yielded a faster cycle time of 3.8 seconds. Next, the effect of implementing 'filling on the move' was investigated. Optimising the acceleration and deceleration phases of the velocity profile as before, it was possible to shorten the cycle time to 3.54 and 3.29 seconds for filling at 10 % and 20 % of the maximum speed respectively.